

AI Team — Main Tasks, Sub Tasks, Allocation

Digital Twin Cyber Deception System (LabyrinthAI) — AI track only Abdelrahman Mohamed · Seif Mohamed · Malak Mahmoud · Martina Saeed

Draft for the team to confirm. Dates follow the official `Task_Assignment.pdf` where they still hold, with September added because Phase 1 finished early.

Where we are

- Chapter 1 submitted to Dr. Sondos on 31 August. All four of us contributed.
- No technical work has started. Nothing is coded, nothing is downloaded.
- `Task_Assignment.pdf` was written on 27 July and never updated. Its Phase 1 starts 1 October, which is after the phase we already finished.
- That leaves September free. It is the only slack we get all year.

One decision blocks everything

Is CSE-CIC-IDS 2018 the right dataset?

- It records **network traffic** — packet counts, byte counts, timings.
- Our classifier is meant to judge attacker skill from **what they do inside a decoy** — commands typed, files opened, how they move.
- Those are different kinds of data. A model trained on one does not transfer to the other.

Three ways out. Pick one before anyone writes code:

- **A** — Keep CSE-CIC-IDS 2018. Accept the classifier is a network intrusion detector. Drop “skill-adaptive” to something smaller and honest.
- **B** — Add command-level data: Cowrie honeypot logs, DARPA OpTC, or the LANL host/network dataset. Two data sources, two models.

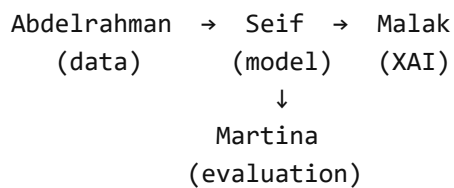
- **C** — Train on our own session logs only. Highest realism, but nothing exists until the Digital Twin runs, so the AI team sits idle until Phase 2 ships.

Owner: Abdelrahman. Needs Dr. Sondos and Eng. Hemdan. Target: 15 September.

Everything below assumes this is settled first.

The chain — why order matters

We are not four parallel workers. We are a queue.



- Seif cannot train until Abdelrahman delivers clean data.
- Malak cannot explain a model that does not exist.
- Martina cannot measure a model that does not exist.

If the front of the chain slips, everyone slips. That is the single biggest risk we carry.

Main Task 1 — Data pipeline

Owner: Abdelrahman · Blocked by: the decision above · Due: 25 Oct 2026

#	Sub task	Due
1.1	Download CSE-CIC-IDS 2018 and verify the files open	20 Sep
1.2	Count real records and class distribution, replacing the estimates in 1.6.1	25 Sep
1.3	Strip identifier columns: IPs, ports, Flow ID, Timestamp	05 Oct
1.4	Remove duplicates, repeated headers, Infinity / NaN	05 Oct
1.5	Build feature engineering: keep behaviour, add the two ratios, cut correlated pairs	15 Oct
1.6	Handle imbalance — undersample, SMOTE, class weights, training split only	20 Oct
1.7	Ship a reproducible script plus a saved scaler Seif can call	25 Oct

Definition of done: Seif can run one command and get train/validation/test sets.

Main Task 2 — LLM decoy generation

Owner: Abdelrahman · **Blocked by:** nothing · **Due:** 27 Dec 2026

#	Sub task	Due
2.1	Install Ollama, run Llama 3 locally, confirm it works offline	20 Sep
2.2	Design the fake company: departments, headcount, names, structure	30 Sep
2.3	Write prompt templates per content type — email, HR file, finance sheet, memo	15 Nov
2.4	Generate a first corpus for one department	30 Nov
2.5	Build the consistency checker: names resolve, file paths resolve, no contradictions	15 Dec
2.6	Screen output for real personal data and credential collisions	20 Dec
2.7	Hand the corpus to the cyber team to load into the decoys	27 Dec

Definition of done: Philopateer explores the twin and cannot tell the content is fake.

This is our most original component. Every commercial deception product uses templates or Faker. LLM-generated linked content is the part that is genuinely current research.

Main Task 3 — Behavioural classifier

Owner: Seif · **Blocked by:** Task 1 · **Due:** 05 Feb 2027

#	Sub task	Due
3.1	Compare architectures: ensemble versus sequence models	25 Oct
3.2	Fix the split — ratio, stratification, seeds, leakage controls	15 Jan
3.3	Train the XGBoost baseline	25 Jan
3.4	Train the LSTM sequence model	05 Feb
3.5	Combine into the hybrid classifier	05 Feb
3.6	Define the skill-tier output the maze controller consumes	05 Feb

Note: 3.2 must match Section 1.6.3. We deleted `Timestamp`, so a time-based split is not available. Agree this with Abdelrahman before writing it.

Main Task 4 — Explainable AI

Owner: Malak · Blocked by: Task 3 · Due: 28 Mar 2027

#	Sub task	Due
4.1	Compare SHAP and LIME for security decisions, pick one, write why	01 Nov
4.2	Integrate onto the trained model	10 Feb
4.3	Define what the analyst sees — which features, how ranked, how worded	28 Feb
4.4	Attach explanations to dashboard events with Mark	28 Mar

Main Task 5 — Evaluation and maze controller

Owner: Martina · Blocked by: Task 3 for evaluation, nothing for the controller · Due: 20 Apr 2027

#	Sub task	Due
5.1	Draft the evaluation metrics framework	01 Nov
5.2	Design the maze controller policy: skill tier in, trap depth out	10 Feb
5.3	Build the test harness — accuracy, precision, recall, F1 scripts	15 Feb
5.4	Add the metrics the allocation names: FPR, engagement time, ATT&CK coverage	28 Feb
5.5	Run the full evaluation with Seif and Malak	15 Apr
5.6	Produce the final performance report	20 Apr

Main Task 6 — Interfaces with the cyber team

Owner: shared · Due: 15 Nov 2026

#	Sub task	Owner	Due
6.1	Agree the log format the SIEM sends us — with Mohamed Salah	Abdelrahman	30 Oct
6.2	Agree how the maze controller triggers containment — with Mohamed Gamal	Martina	15 Nov
6.3	Agree the dashboard contract — with Mark	Malak	15 Nov

Why this exists. Chapter 1 is written but no two people have agreed an actual data format. This is where integration projects usually break.

Main Task 7 — Final report

Owner: shared · Due: 10 May 2027

#	Sub task	Owner
7.1	AI and methodology sections	Abdelrahman + Seif
7.2	Evaluation and XAI sections	Malak + Martina

This week

Task	Owner
Take the dataset question to Dr. Sondos and Eng. Hemdan	Abdelrahman
Fix the TOC bug in <code>Phase 1.docx</code> — the decoy bullets are styled as headings	Abdelrahman
Download CSE-CIC-IDS 2018	Abdelrahman
Install Ollama and Llama 3	Abdelrahman
Start the SHAP versus LIME comparison	Malak
Start the metrics framework draft	Martina
Start the architecture comparison	Seif

Malak, Martina and Seif can all start their research tasks now. None of them need data.

For Kareem

- **Tooling.** Eng. Ramez said on 19 August that Notion is no good and pushed ClickUp or Trello. Decide before we build the board twice.
- **Ownership clash.** `Task_Assignment.pdf` makes Martina Evaluation Lead. The WhatsApp split gave Section 1.6.6 Evaluation Metrics to Seif. One of the two is wrong.
- **Name.** There is a commercial deception platform called Labyrinth. Worth settling before anyone presents.
- **Name on file.** The allocation lists Martina Saeed. She corrected this in the group on 27 July to Martina Khalil. It is on a document going to the Dean.